

Stock span

arr = [100, 80, 60, 70, 60, 75, 85]

output = [1, 1, 1, 2, 3, 4, 6]

Ans:- Nearest greater Integer find Karte hai.

arr = [100, 80, 60, 70, 60, 75, 85]

index

0 - (-1) = -1

0 - (-1) = 1

1 - 0 = 1

2 - 1 = 1

3 - 1 = 2

4 - 3 = 1

5 - 4 = 1

6 - 5 = 1

output

→ [1, 1, 1, 2, 3, 4, 6]

→ [i - index]

→ Same as greater next Integer from left

→ Pehle next greater find karo.

→ Phir uska index pata karo.

→ last me $i - \text{index}$ karalo.

// Code:

```
public static int[] stockSpan(int[] prices) {  
    int n = prices.length;  
    int[] span = new int[n];
```

```
    Stack<Integer> stack = new Stack<>();
```

```
    for (int i = 0; i < n; i++) {
```

```
        while (!stack.isEmpty() && prices [stack  
            prices[stack.peek()] <= prices[i]) {
```

```
            stack = pop();
```

```
        }
```

```
        if (stack.isEmpty()) {
```

```
            span[i] = i + 1;
```

```
        } else {
```

```
            span[i] = i - stack.peek();
```

```
        }
```

```
        stack.push(i);
```

```
    }
```

```
    return span;
```

```
}
```

```
}
```